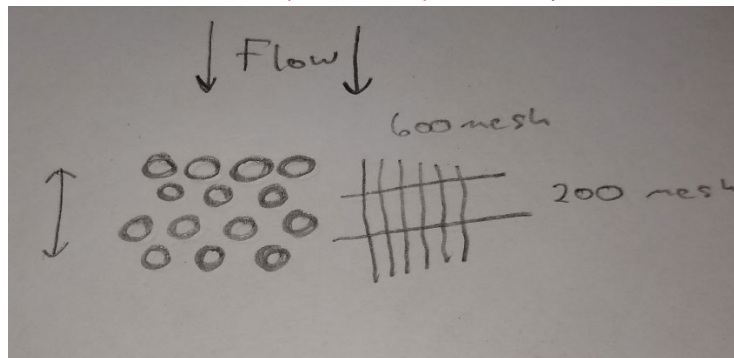


Purchase specifications:

The 9x15 wind tunnel requires additional face sheets for future acoustic **box** construction. These boxes are the acoustic treatment that line the new 9x15 wind tunnel test section and provides the anechoic properties as well as smooth surface characteristics required **for** the sensitive acoustic testing done in the 9x15. Due to the sensitivity of the testing, the tolerances in manufacturing must be held extremely tightly since the cumulative effect of every defect will combine to cause errors in measurement. This will impede testing and limit the tunnels capabilities. The panels **consist a woven wire mesh diffusion bonded to a perforated plate. The perforated plate is a 304l stainless steel plate with 63% open area and 5/16" staggered round holes** of various listed thickness. **The woven wire mesh is a 200x600 304l stainless steel Dutch Twill woven wire mesh.** The sheets are **either 16 or 20 gauge** thickness with **wire mesh oriented as specified in the sketch below.** This design has been **finalized after hundreds of hours of testing in the Nasa Glenn ATL and at the Virginia Polytechnic Institute over about two years of research.** Any new manufacturer will require similar levels of validation of sample products that can only be conducted through a rigorous testing regiment. Therefore, due to the amount of effort expended in validating the existing supplier, we feel that a sole source justification **is** required since all existing suppliers required significant support and testing to refine their processes to meet our specification. This would come at a great cost of manpower and resources to reproduce.

Requirements

1. Quantity required is
 - a. 25 panels (48.5" x 48.5") minimum in 20 gauge panels
 - b. 25 panels (48.5" x 48.5") minimum in 16 gauge panels
2. Perforated panels must be of a staggered round hole orientation with a 63% open area with screen bonded to machine finished side for maximum smoothness
3. Wire mesh screen must have a wire thickness of .002" (200 mesh) x.0018" (600 mesh)
4. Screen must be diffusion bonded to perforated **plate with the 600-mesh wires aligned with the staggered hole direction of the perforated plate as in pictured below.**



- a.
5. Diffusion bonded panel must be free of all contaminants and not have any residual material left over by bonding process. Particularly in the wire mesh portion.
6. Diffusion bonding process may not result in screen dimples **with a depth greater than 0.007 inch** into **the** perforate.

7. Wrinkled, ripped, creased and punctured panels shall be rejected.
8. Finished panels must have a maximum acoustic permeability of 20 cgs Rayls per ASTM C522.
9. Overall panel flatness must be ¼ inch and less than 1/16 inch per foot.
10. The wire cloth shall not peel from the perforated plate when pulled with a force up to 15 lbs.
11. The wire mesh shall cover the entire perforated plate.
12. The finished panels shall be free of burn marks and discoloration.

Attached is an existing quote from our primary OEM supplier (PMF) who has previously shown capability to meet our requirements.